
Reaction Coordinates for Specification-Conditioned Neural Fields

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Abstract

A parametric physics-informed surrogate cannot respond to a changed initial condition unless that condition is part of its input. Whether a structured description improves generalization beyond direct samples is a separate question. We study Fourier descriptions of initial profiles in raw and log-odds coordinates while holding a reference neural field, decoder, physics loss, and training budget fixed. Logistic reaction motivates the nonlinear coordinates and supplies exact evaluation references. Across three seeds and 151 registered queries, reaction-coordinate conditioning reduces relative-L2 error on Gaussian interpolation from 0.0740 to 0.0461 and on sinusoidal profiles from 0.0635 to 0.0519, while retaining original-profile accuracy. However, polynomial transfer and unseen pairings of coefficients and profiles do not satisfy the registered seed-variation criterion. Conditioning removal sharply degrades accuracy, and distance from the training descriptor span tracks extrapolation error. An exact weight conversion also shows that ordinary Fourier conditioning changes parameterization rather than expressivity. These results distinguish useful physics-aligned interpolation from a broader generalization guarantee: a meaningful specification code does not, by itself, establish a transferable solution operator.

1 Conditioning on coefficients and initial profiles

A surrogate for a differential equation is useful only if its inputs identify the problem being solved. A reaction rate does not identify a solution when the initial concentration profile can change. Physics-informed neural networks (PINNs) impose an initial profile through a loss [Raissi et al., 2019]; coefficient-conditioned surrogates amortize that optimization across coefficients [Cho et al., 2024]. Making the profile an additional input removes this ambiguity, but introduces a harder question: does an encoder learn a transferable description of a function, or just distinguish a small collection of training profiles?

We examine this question through an intentionally controlled nonlinear problem. The solution map for logistic reaction is nonlinear in the initial condition, yet an exact reference is available for every query. The architecture outside the specification encoder is fixed. This lets us distinguish three explanations for an apparent improvement: access to the initial condition, a change of basis in its representation, and a nonlinear coordinate system suggested by the governing equation. These explanations are often conflated when the decoder, training objective, and conditioning pathway change together.

Our candidate representation applies a Fourier transform to the initial condition in log-odds coordinates. Logistic reaction translates these coordinates uniformly in time, providing a concrete physical rationale for the representation. Crucially, the network does not receive the solution, transformed time evolution, or a hard-coded solution ansatz. It receives only a finite description of the initial profile and the coefficients; the original neural field still learns the solution over space and time through the original physics loss.

The study compares coefficient conditioning, direct sensor conditioning, ordinary Fourier conditioning, and reaction-coordinate Fourier conditioning. It separates interpolation of Gaussian parameters,

interpolation of reaction rates, new pairings of familiar coefficients and profiles, and transfer to sinusoidal and polynomial profiles. Input interventions test whether conditioning is used; latent diagnostics test what that dependence establishes. We also distinguish an invertible representation from a learned extension away from the training profiles. The resulting evidence concerns a restricted nonlinear family, rather than a general-purpose operator solver.

2 A specification encoder with a fixed neural field

Problem and architecture. We study the reaction member of the convection-diffusion-reaction family,

$$\mathcal{R}_\mu[u] = u_t + \beta u_x - \nu u_{xx} - \rho u(1 - u), \quad \mu = (0, 0, \rho), \quad x \in [0, 2\pi), \quad t \in [0, 1]. \quad (1)$$

The initial profile is $u(x, 0) = a(x)$, with periodic endpoint values. A specification is $s = (\mu, a, b)$, where b describes the boundary type. Only the periodic value $b = 1$ occurs in this study. It is therefore not an experiment on boundary-condition generalization.

We import the reference P2INN_phase1 implementation [Cho et al., 2024]. Its coordinate encoder $c_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^{50}$, residual decoder, and concatenation of parameter and coordinate embeddings are unchanged. The specification encoder replaces only the input to the parameter branch:

$$h_s = E_\theta([\mu; \psi(a); b]) \in \mathbb{R}^{50}, \quad \hat{u}(x, t; s) = D_\theta([h_s; c_\theta(x, t)], \mu, x, t). \quad (2)$$

The second argument of D_θ makes explicit the reference decoder’s existing modvec shift: a linear map of $[\mu; x; t]$ is added before its final hidden activation. No IC descriptor enters that pathway. All components train jointly.

Three descriptions of the same sensors. We evaluate a at $M = 32$ equally spaced midpoint sensors $x_j = 2\pi(j + 1/2)/M$, obtaining $y_j = a(x_j)$. The sensor-conditioned model uses $\psi_{\text{raw}}(a) = y$. Let $c_k = M^{-1} \sum_j y_j \exp(-2\pi i j k / M)$. Define the real transform

$$\mathcal{F}_R(y) = (c_0, \sqrt{2}\Re c_{1:15}, \sqrt{2}\Im c_{1:15}, c_{16}) \in \mathbb{R}^{32}. \quad (3)$$

This retains every discrete mode, with $\|\mathcal{F}_R(y)\|_2^2 = M^{-1}\|y\|_2^2$. Ordinary Fourier conditioning uses $\psi_F = \mathcal{F}_R(y)$. Reaction Fourier conditioning uses

$$\psi_R(a) = \mathcal{F}_R\left(\frac{1}{4} \log \frac{\tilde{y}}{1 - \tilde{y}}\right), \quad \tilde{y} = \min(1 - \epsilon, \max(\epsilon, y)), \quad \epsilon = 10^{-4}. \quad (4)$$

The scale and clipping threshold are fixed before training. There is no spectral truncation, learned preprocessing, or positive-time reference in these inputs. The coefficient-only model receives just μ .

Each conditioned branch has dimensions $36 \rightarrow 150 \rightarrow 150 \rightarrow 150 \rightarrow 50$, with tanh between its hidden layers. These branches all have 58,400 parameters, and each complete model has 81,801. The coefficient-only model has 53,450 encoder parameters and 76,851 total parameters. All methods share the same 5,250-parameter coordinate architecture and 18,151-parameter decoder including its shift. Identical seeds initialize the shared components identically.

Why reaction coordinates are plausible. For $0 < a(x) < 1$, set $q(x, t) = \log(u(x, t)/(1 - u(x, t)))$. Equation (1) gives

$$q_t = \frac{u_t}{u(1 - u)} = \rho, \quad q(x, t) = q(x, 0) + \rho t. \quad (5)$$

Thus only the constant Fourier mode of q changes with time. This identity motivates the input representation; it does not replace the decoder or its objective. In particular, the architecture does not enforce this translation law.

The logistic inverse $\sigma(q)$ is 1/4-Lipschitz. Consequently, two unclipped profiles in reaction coordinates obey

$$\|\sigma(q_0 + \rho t) - \sigma(\bar{q}_0 + \rho t)\|_2 \leq \frac{1}{4} \|q_0 - \bar{q}_0\|_2. \quad (6)$$

This is an absolute representation bound, not a relative-error guarantee for a fitted network. It does not bound learned decoder error, guarantee sampling resolution, or remove clipping error. Near zero and one, the forward log-odds map is sensitive, so a physically simple coordinate system need not be statistically easy to learn.

An important equivalence. Because $\mathcal{F}_R$ is invertible and the first encoder layer is affine, raw and ordinary Fourier conditioning have exactly the same representable functions on the sensors. If row-vector descriptors satisfy $z = yB$, a raw first-layer IC weight W is represented in Fourier coordinates by $WB^{-\top}$. Any observed difference between these arms concerns parameterization, initialization, and finite-budget optimization, not a larger hypothesis class. The nonlinear map in Equation (4) changes this parameterization more substantially; clipping can additionally lose information.

3 A registered evaluation across specifications

Profiles and splits. Training uses centered Gaussians $G(A, w; x) = A \exp[-(x - \pi)^2 / (2\pi^2 w^2)]$ and integer reaction rates $\rho \in \{1, 2, 3, 4, 5\}$. Five profiles have (A, w) equal to $(1, 1/2)$, $(1/2, 1/2)$, $(1, 1/4)$, $(1/2, 1/4)$, and $(3/4, 3/8)$, denoted G0 through G4. At each rate we exclude profile $1 + ((\rho - 1) \bmod 4)$, leaving twenty training pairs and five unobserved pairings. G0, the original `gauss_pi_2` profile, is present at every training rate.

Gaussian interpolation uses $(A, w) \in \{5/8, 7/8\} \times \{5/16, 7/16\}$. Shape transfer uses two sinusoids, $0.55 - 0.35 \cos x$ and $0.50 - 0.45 \cos x$, and two polynomial profiles, $0.10 + 0.80(1 - r^2)^2$ and $0.20 + 0.50(1 - r^2)^2$, where $r = (x - \pi)/\pi$. None of these profiles is trained. The new profiles have compatible periodic endpoint values; the polynomials also have equal endpoint derivatives. All profiles are nonnegative and bounded by one, avoiding reaction blow-up for negative initial values.

Coefficient interpolation uses $\rho \in \{1.5, 2.5, 3.5, 4.5\}$ with G0 through G4. Joint interpolation pairs these same rates with the four new Gaussian profiles. A further sweep fixes $A = 0.75$, varies $w \in \{0.1, 0.15, 0.2, 0.25, 0.3, 0.375, 0.4, 0.5, 0.6, 0.75\}$ and uses $\rho \in \{1, 3, 5, 7, 10\}$. This diagnostic deliberately revisits some training profiles and extends to narrower, broader, and faster-reaction problems. It is not counted as an entirely unseen set. In total the registration contains 151 evaluation specifications, including twenty seen pairs. Every registered specification is reported in the accompanying data.

Budget and loss. Each training pair uses the reference 1,000 interior points, 256 initial points, and 100 periodic endpoint pairs. The coefficient-only arm uses G0 in all twenty cells, with four identical copies per rate. Repetition matches executed point counts; because the coordinates coincide and losses are means, its objective equals the original five-rate, fixed-profile objective. It receives no additional distinct collocation data.

All arms minimize the unchanged objective

$$L = \underbrace{\langle \mathcal{R}_\mu[\hat{u}]^2 \rangle_f}_{L_f} + \underbrace{\langle \mathcal{R}_\mu[\hat{u}]^2 \rangle_{t=0}}_{L_u} + \underbrace{\langle (\hat{u}(x, 0) - a(x))^2 \rangle}_{L_{IC}} + \underbrace{\langle (\hat{u}(0, t) - \hat{u}(2\pi, t))^2 \rangle}_{L_b}. \quad (7)$$

All four weights are one. We call the reference residual routine, including its spatial derivatives even when their coefficients are zero. Adam uses learning rate 10^{-3} and its default scalar settings for exactly 20,000 updates with seeds 0, 1, and 2. Each point set fits in a single reference-sized batch; all points contribute at every update. Float-valued specification plumbing avoids the reference loader’s integer casts. Validation coordinates from seen pairs are monitored every 5,000 updates. No hyperparameters, checkpoints, or training budgets are selected from validation or test errors; all tables use the final update.

References and metrics. The spatial and temporal reference grids remain 256×100 on the original domain. We call the unmodified `systems.reaction_solution`; its function interface already accepts callable profiles. For reaction,

$$u(x, t) = \frac{a(x)e^{\rho t}}{1 - a(x) + a(x)e^{\rho t}}. \quad (8)$$

The two corresponding zoo strings agree with the callable Gaussians. Refinement to 512×199 aligns every coarse grid point; the reference is analytic, so this checks implementation consistency rather than convergence of a numerical solver. No non-periodic or operator-split reference is used.

We retain the reference sampler’s fixed, disjoint sets of 1,000 interior training, test, and validation points. Primary error is $\|\hat{u} - u\|_2 / \|u\|_2$, computed separately per specification. The source implementation calls mean absolute error “absolute L2”; we report it explicitly as MAE and also provide

Table 1: Reaction-family results at identical point counts and 20,000 Adam updates. Each cell gives relative L2 (top) and MAE (bottom), both as mean $\pm$ sample SD over three seeds. “Original profile” is G0 at the five training rates; “training specifications” contains all twenty trained pairs. Other rows are disjoint primary test sets. The coefficient-only arm always predicts its fixed-profile solution. True absolute L2 norms appear in Table 6.

Query set	Coefficient only	Raw sensors	Fourier	Reaction Fourier
Original profile	0.0035 $\pm$ 0.0021	0.0049 $\pm$ 0.0015	0.0036 $\pm$ 0.0016	0.0041 $\pm$ 0.0011
	0.0024 $\pm$ 0.0017	0.0029 $\pm$ 0.0008	0.0022 $\pm$ 0.0010	0.0027 $\pm$ 0.0010
Training specifications	0.5040 $\pm$ 0.0032	0.0183 $\pm$ 0.0039	0.0207 $\pm$ 0.0021	0.0177 $\pm$ 0.0052
	0.2303 $\pm$ 0.0020	0.0064 $\pm$ 0.0015	0.0067 $\pm$ 0.0011	0.0064 $\pm$ 0.0023
Gaussian interpolation	0.4140 $\pm$ 0.0033	0.0740 $\pm$ 0.0048	0.0628 $\pm$ 0.0197	0.0461 $\pm$ 0.0053
	0.2163 $\pm$ 0.0022	0.0357 $\pm$ 0.0024	0.0302 $\pm$ 0.0091	0.0216 $\pm$ 0.0030
Sinusoidal profiles	0.1039 $\pm$ 0.0019	0.0635 $\pm$ 0.0053	0.0662 $\pm$ 0.0037	0.0519 $\pm$ 0.0064
	0.0609 $\pm$ 0.0018	0.0353 $\pm$ 0.0021	0.0353 $\pm$ 0.0029	0.0281 $\pm$ 0.0054
Polynomial profiles	0.1071 $\pm$ 0.0022	0.0834 $\pm$ 0.0004	0.0670 $\pm$ 0.0157	0.0719 $\pm$ 0.0178
	0.0626 $\pm$ 0.0018	0.0475 $\pm$ 0.0017	0.0380 $\pm$ 0.0095	0.0413 $\pm$ 0.0115
Recombined pairs	0.6107 $\pm$ 0.0037	0.1167 $\pm$ 0.0261	0.1333 $\pm$ 0.0641	0.0987 $\pm$ 0.0211
	0.2821 $\pm$ 0.0022	0.0500 $\pm$ 0.0104	0.0568 $\pm$ 0.0266	0.0430 $\pm$ 0.0119
Coefficient interpolation	0.5109 $\pm$ 0.0055	0.0646 $\pm$ 0.0254	0.0778 $\pm$ 0.0456	0.0508 $\pm$ 0.0177
	0.2397 $\pm$ 0.0017	0.0274 $\pm$ 0.0107	0.0330 $\pm$ 0.0197	0.0220 $\pm$ 0.0090
Joint interpolation	0.3991 $\pm$ 0.0071	0.0813 $\pm$ 0.0107	0.0776 $\pm$ 0.0357	0.0570 $\pm$ 0.0187
	0.2126 $\pm$ 0.0034	0.0384 $\pm$ 0.0036	0.0373 $\pm$ 0.0172	0.0271 $\pm$ 0.0096

the true Euclidean absolute norm. We average specifications within each seed, then report mean and sample standard deviation across seeds. A coefficient-only prediction on a changed profile is scored against that query’s true solution: it represents the reachable prediction of a model that cannot accept the change, not retraining at the new profile.

Decisions and interventions. Before comparison runs, we registered a minimum improvement $\delta = 0.005$ in relative-L2 units against both coefficient-only and raw-sensor conditioning, and required a margin above their pooled seed standard deviation. Original-profile accuracy is checked at each coefficient, allowing an increase of at most $\max(0.1e_{\text{base}}, 0.005)$. These are descriptive decision rules; three seeds do not justify strong population-level significance claims. Full paired differences and paired t intervals accompany the results.

At evaluation, constant-profile conditioning replaces only the IC descriptor by G0; a second intervention cyclically permutes profiles within each query set while preserving coefficients. Latent diagnostics use a fixed rate, training-profile principal components, leave-one-profile-out linear probes, and a distance-to-training analysis. These interventions and analyses change no trained weights.

4 What transfers, and what remains uncertain

Initial-condition access is necessary, but structure has a smaller effect. Table 1 reports every primary query axis. On the original profile, the coefficient-only implementation reaches relative error 0.0035 ± 0.0021 , consistent in scale with the source paper’s small-error reaction results. Once profiles vary, its error is much larger because its output cannot depend on that variation. All three conditioned models reduce this ambiguity. Their differences therefore measure the additional benefit of a representation, beyond simply exposing the profile to the network.

Reaction Fourier conditioning reduces Gaussian interpolation error from 0.0740 to 0.0461, and sinusoidal-transfer error from 0.0635 to 0.0519. Simultaneously interpolating the coefficient and Gaussian profile gives 0.0570 ± 0.0187 , compared with 0.0813 ± 0.0107 for direct sensors. These three mean improvements exceed both the registered 0.005 floor and pooled seed SD (Table 2). The sinusoidal result is transfer beyond the Gaussian training family, but only to the two registered sinusoidal profiles; it is not evidence for arbitrary functions.

Table 2: Reaction Fourier improvements in relative-L2 units. Positive gains favor the method. A comparison satisfies the registered rule only when gains against both reference arms exceed δ and their respective pooled seed SDs. Passing this rule is not a claim of statistical significance.

Query set	Gain vs. raw	Pooled SD	Gain vs. coefficient	Decision
Gaussian interpolation	0.0279	0.0050	0.3679	Pass
Sinusoidal profiles	0.0116	0.0058	0.0520	Pass
Polynomial profiles	0.0115	0.0126	0.0351	Fail
Recombined pairs	0.0180	0.0237	0.5120	Fail
Joint interpolation	0.0243	0.0153	0.3421	Pass

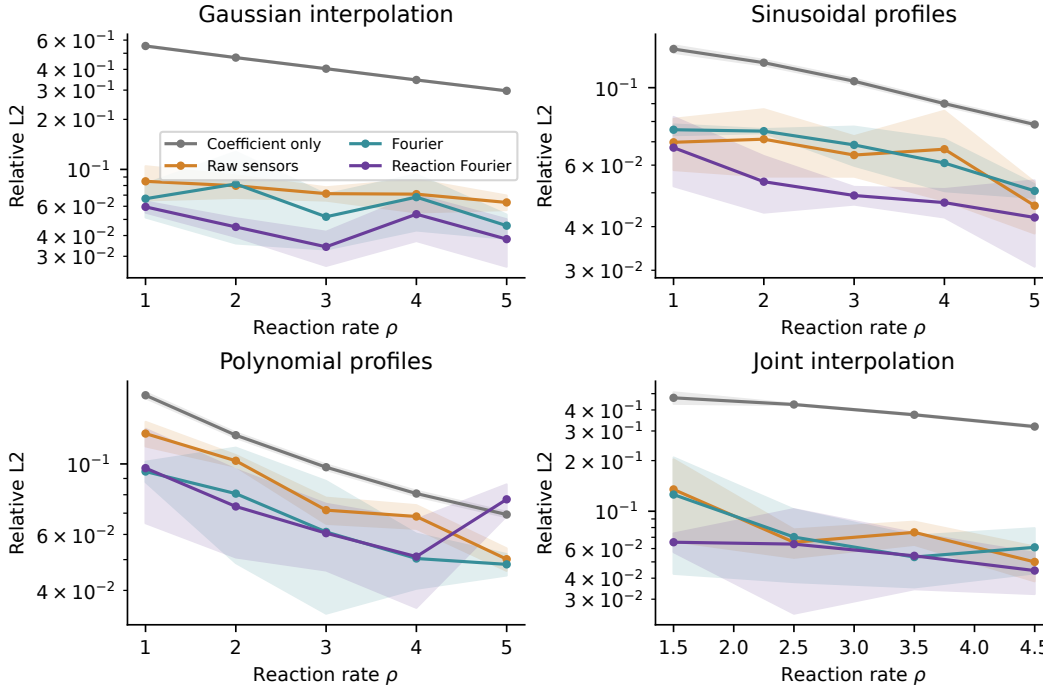


Figure 1: Error versus reaction rate on four transfer axes. Lines average all profiles on the indicated axis within a seed, then average seeds; shading is one seed SD. Reaction-coordinate conditioning improves several means, but neither profile-family transfer nor joint interpolation is uniformly favorable at every rate.

The broader superiority hypothesis is not established. Polynomial transfer improves in the mean, but its raw-sensor margin is smaller than the pooled seed SD. The same is true for the five unobserved pairings of known profiles and integer rates. Figure 1 shows that relative performance also varies with rate. Ordinary Fourier conditioning sometimes improves the mean, especially for polynomials, but is not uniformly superior either. We retain all of these results instead of selecting the favorable representation separately for each query set.

The uncertainty is material even for comparisons that satisfy the registered rule. Only Gaussian interpolation has a positive lower endpoint in the paired 95% t interval against raw sensors; the sinusoidal and joint-interpolation intervals include zero. One joint-interpolation seed reverses the ordering. With three seeds and a small, fixed profile collection, these data support a bounded empirical improvement, not a universal or high-confidence population claim. Appendix C reports the paired intervals and individual-query curves.

The original capability is retained. Reaction Fourier gives 0.0041 ± 0.0011 on G0. All five per-coefficient mean increases remain within the registered non-regression tolerance; the raw and

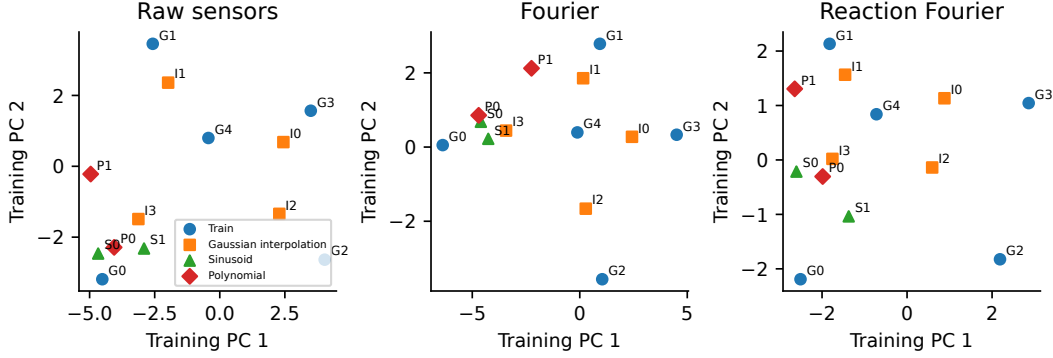


Figure 2: Initial-profile latents at a fixed reaction rate, using seed 0 and principal directions fitted to the five training codes. G denotes training Gaussians, I new Gaussian parameters, S sinusoidal profiles, and P polynomial profiles. Projected proximity does not exclude displacement outside the displayed plane; predictive errors are evaluated separately.

Table 3: Representation diagnostics, mean $\pm$ seed SD. Rank uses the five training profiles; the linear width probe excludes each profile in turn. Correlations compare ten width-sweep errors at $\rho = 3$ with either distance to the nearest training profile or distance to the affine span of training descriptors. They are descriptive correlations, not independent replications or calibrated error bounds.

Encoder	Latent rank	Width probe R^2	IC-distance corr.	Span-distance corr.
Raw sensors	2.523 ± 0.147	0.470 ± 0.421	0.467 ± 0.111	0.838 ± 0.092
Fourier	2.007 ± 0.032	0.592 ± 0.138	0.515 ± 0.128	0.806 ± 0.032
Reaction Fourier	2.129 ± 0.140	0.738 ± 0.167	0.584 ± 0.050	0.895 ± 0.031

ordinary Fourier arms also pass. This check is separate from the broader training-pair average and prevents a large gain on changed profiles from hiding a loss of original-profile accuracy. Table 5 gives the complete per-coefficient comparison.

5 Conditioning, geometry, and the limits of the mechanism

The network uses the initial-condition input. Replacing the Reaction Fourier IC descriptor by G0 raises Gaussian interpolation error from 0.0461 to 0.4100 ± 0.0047 ; cyclic profile permutation gives 0.3060 ± 0.0159 . Sinusoidal-transfer error rises to 0.1016 ± 0.0026 under the constant intervention. The degradation exceeds the registered noise rule on every primary axis for both interventions. Raw and ordinary Fourier models also deteriorate (Table 8), so this test establishes use of conditioning rather than uniquely validating the proposed representation.

At $\rho = 3$, the Reaction Fourier latent distinguishes all five training profiles: its mean within-model pairwise distance is 3.887 ± 0.295 and its smallest pairwise distance is 2.196 ± 0.420 . The coefficient-only latent has zero IC-dependent distance. Latent scales are not calibrated across independently trained models, so a larger distance is not automatically better.

A compact geometry is compatible with learning profile parameters. Figure 2 projects each model onto principal components fitted only to its training-profile codes. Table 3 summarizes the effective rank, defined from normalized squared singular values, and leave-one-profile-out probes. The Reaction Fourier width probe has $R^2 = 0.738 \pm 0.167$, versus 0.470 ± 0.421 for raw sensors. The amplitude probe is much less stable: Reaction Fourier has $R^2 = 0.515 \pm 0.689$, including a negative score in one seed. With only five training profiles, these probes are diagnostic evidence of geometric organization, not a reliable estimator of parameter disentanglement.

Leaving the training descriptor span tracks error. For each encoding, we compute the distance of a query descriptor to the affine span of the five training descriptors. This requires the IC alone and does not fit a predictor to test errors. Across the width sweep at $\rho = 3$, this distance has

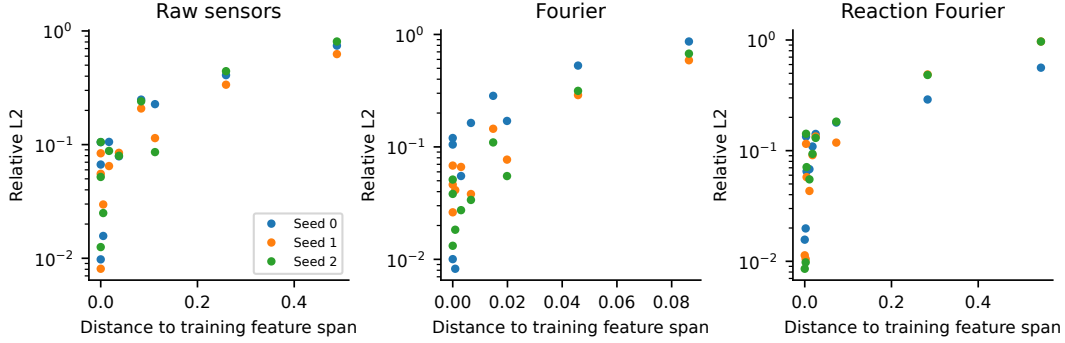


Figure 3: Descriptor-span distance versus error for the fixed-rate width sweep. Each marker is one specification and seed. Span geometry tracks the growth of error for all three conditioned encoders, including failures of Reaction Fourier outside the training region.

Spearman correlation 0.895 ± 0.031 with Reaction Fourier error, compared with 0.584 ± 0.050 for nearest-profile distance in sensor space (Figure 3). Similar correlations occur for the other encoders. The result supports a shared explanation: the models organize the training profiles, but extrapolating beyond their descriptor geometry remains difficult. It does not demonstrate that the hidden state implements the exact translation law in Equation (5).

Fourier coordinates do not create extra expressivity. We also transform each trained raw-sensor model’s first-layer weights by the exact inverse basis map, without optimization. Its predictions then agree with the equivalent Fourier-input model over every registered query with maximum discrepancy 2.554×10^{-15} in float64. This is an algebraic diagnostic, distinct from the independently trained Fourier arm in Table 1. It confirms that any advantage of that arm comes from the training parameterization. Reaction coordinates provide a more appropriate description of logistic dynamics, but neither their invertibility nor their physical motivation forces the learned decoder to extrapolate correctly.

6 Extrapolation exposes a remaining limitation

Figure 4 varies Gaussian width on both sides of the training range. Narrow profiles remain difficult for every model, and Reaction Fourier is not the most accurate arm across the full width/rate sweep: its relative error is 0.3052 ± 0.0201 versus 0.2380 ± 0.0245 for ordinary Fourier and 0.2758 ± 0.0254 for raw sensors. Thus an interpolation gain does not imply a more reliable extrapolator. The sweep includes coefficient extrapolation as well as width changes; Table 9 separates representative widths and rates.

The reaction reference is analytic and consistent under aligned grid refinement, so these failures are not errors from a time-stepping reference. The nonlinear feature map does clip sufficiently small tails. A separate exploratory calculation propagates the clipped initial profile through the exact reference and measures this distortion (Table 9). At moderate reaction rates it is far smaller than the learned errors, so clipping alone cannot explain the observed failure. At faster reaction rates, amplification of small tails makes clipping more consequential. Sparse profile coverage, finite decoder training, and the conditioning geometry remain competing explanations. We did not expand encoder width, collocation, or training duration; these are finite-budget failures, not established fundamental boundaries.

7 Relation to function-conditioned surrogates

Conditioning a physics-informed model on a class of problems is established: conditional PINNs address parametric eigenvalue problems [Kovacs et al., 2022], and P²INNs explicitly encode coefficients before decoding a coordinate-dependent solution [Cho et al., 2024]. DeepONet separates a function-encoding branch from a coordinate-encoding trunk [Lu et al., 2021]; physics-informed DeepONets train such mappings with differential constraints and specified initial or boundary data

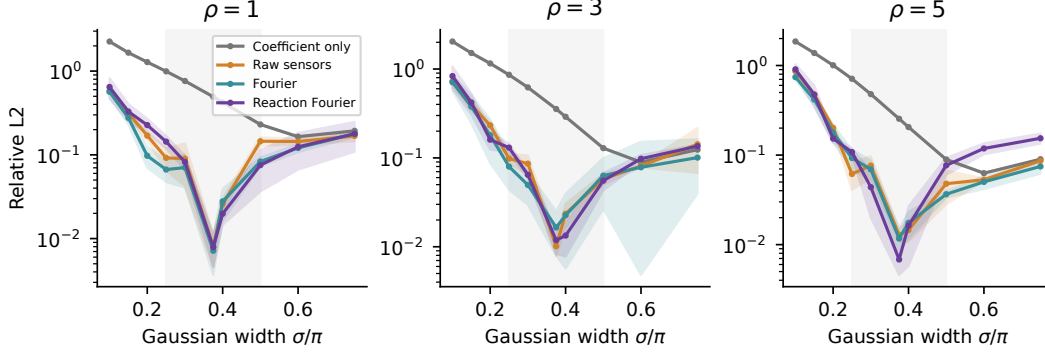


Figure 4: Width extrapolation at fixed reaction rates. Gray shading marks the training width interval, not a densely observed training region. Only five training profiles are available. Error grows for narrow profiles, and the proposed nonlinear coordinates do not dominate either alternative outside the training region. Shading around curves is one seed SD.

[Wang et al., 2021]. We therefore do not claim that exposing initial conditions to a PINN is itself new. Our comparison isolates representations at a fixed concatenation interface and asks whether physics-motivated coordinates improve transfer beyond direct sensors.

Fourier neural operators learn integral operators with Fourier-parameterized kernels [Li et al., 2021]. Here Fourier analysis transforms only the initial specification; the decoder is an ordinary coordinate field and the transform retains the entire finite sensor vector. Functa represents observations as functions with compact neural representations [Dupont et al., 2022]; CORAL extends coordinate-based operator learning to general geometries [Serrano et al., 2023]. Our fixed periodic interval, finite sensor grid, and restricted reaction family do not address that geometric breadth. The exact logistic coordinate change is classical algebra; the empirical contribution is testing what that coordinate change achieves when it is confined to the specification encoder. Known PINN optimization failures [Krishnapriyan et al., 2021] also motivate our distinction between a finite-budget transfer failure and a fundamental representational obstruction.

8 Scope and implications

The evaluation spans three IC families, one boundary type, and one PDE family; only the Gaussian IC family is trained. It establishes use of the IC input and several improvements under a fixed computation budget, while leaving superiority on polynomial profiles and unseen pairings unresolved. There is no demonstrated boundary-condition transfer, joint support for multiple PDE families, or transfer to a different differential operator.

Reaction is nonlinear in the initial profile but spatially decoupled. This makes the experiment diagnostic and the references trustworthy, while limiting its relevance to transport and diffusion. All profiles are centered and smooth, the function-family tests contain only two profiles each, and the width probe has only five training examples. The fixed sensor representation also lacks resolution independence. The reaction-coordinate transform uses a special analytic property that need not exist for another operator. No accuracy or speed advantage over the exact logistic solver is asserted.

The most useful distinction is between representing the specification and learning its extension. Input interventions and distinct codes establish the first; they do not establish the second. Physically motivated coordinates improve several interpolation means without guaranteeing compositional or extrapolative accuracy. The ordinary Fourier weight-conversion experiment makes the same point from another direction: an appealing representation can preserve the hypothesis class exactly. Future work would need richer profile coverage and independent families before treating the observed organization as evidence of a general solution operator.

9 Conclusion

A fixed neural-field architecture can be conditioned on changing initial profiles without sacrificing its original fixed-profile accuracy. For nonlinear reaction, log-odds Fourier conditioning improves Gaussian interpolation and selected shape-transfer means relative to equally sized sensor conditioning. Yet the complete registered superiority hypothesis is not established: polynomial transfer and unseen pairings remain sensitive to seed variation, and extrapolation can favor simpler inputs. A specification encoder’s physical motivation, distinguishable latent codes, and good interpolation should therefore be evaluated separately from its ability to generalize across the full problem specification.

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A Implementation, reference checks, and reproducibility

The delivered code imports the supplied reference model and inherits its forward function verbatim. A pre-hook to the first parameter layer supplies one descriptor per specification; a hook to the final parameter layer expands its output to the coordinate rows. This computes repeated embeddings once without changing the model. The coordinate layers and all decoder layers, including `modvec`, remain the original objects. A dense version that repeats descriptors before encoding agrees with this batched version in output, residual, loss, and parameter gradients. The maximum float64 gradient discrepancy is 2.220×10^{-16} . A direct coefficient-only comparison against an unmodified reference instance also agrees to floating-point precision. These checks are in `logs/verification.json`.

Three warm-up optimizer steps initialize CUDA execution. Graph capture records, but does not execute, the update; 19,997 replays then complete the 20,000-step budget. Every optimizer state counter is checked. A completed implementation check detected a one-update counting error; all four corrected 100-update checks subsequently reached exactly 100 optimizer steps. These preliminary runs are excluded from the scientific comparisons, and their measured outcomes are documented in `attempts_log.md`. Independent seeds ran on three available NVIDIA H200 devices, one device per model, with float32 arithmetic and TF32 disabled. In total the twelve final-epoch models used 240,000 Adam updates and 34.4 minutes of summed model training time, including scheduled seen-specification validation and checkpoint writes. Evaluation and engineering checks are not included in that timing. GPU distribution changes wall-clock throughput, not the per-model scientific budget.

The two original Gaussian string references and callable references agree exactly. All registered reaction references agree exactly at coinciding coarse and refined grid points. An independent float64 automatic-differentiation check of the logistic formula has maximum reaction residual 2.109×10^{-15} . These are numerical consistency checks for an analytic reference, not a convergence claim for a new solver. The real Fourier transformation has condition number one up to roundoff; the maximum all-query prediction discrepancy under the raw-to-Fourier first-layer weight conversion is 2.554×10^{-15} . That conversion uses no optimizer updates.

The timestamped plan and its hash are stored in `code/specification_encoder_plan.json` and `logs/registration.json`; each training manifest records that same hash and source hashes. `logs/per_specification.csv` contains every specification, seed, method, intervention, and error metric. Prediction arrays, references, final checkpoints, training histories, and latent coordinates are retained. Tables and figures are generated from these records by `code/analyze.py` and `code/supplement.py`. The primary training script does not access the evaluation specification list. No supplied pretrained checkpoint is used. No external methodology implementation is copied.

B Detailed split accounting

Table 4: Registered specifications. Each rate is paired with every listed profile except the explicit recombination exclusions. The diagnostic sweep revisits some training combinations and is labeled separately.

Set	Profile/rate definition	Count
Training	G0 through G4, rates 1 to 5, excluding the five pairs below	20
Recombined pairs	(1, G1), (2, G2), (3, G3), (4, G4), (5, G1)	5
Gaussian interpolation	Four (A, w) pairs, each at rates 1 to 5	20
Sinusoidal profiles	S0 and S1, each at rates 1 to 5	10
Polynomial profiles	P0 and P1, each at rates 1 to 5	10
Coefficient interpolation	G0 through G4 at four half-integer rates	20
Joint interpolation	Four new Gaussian profiles at four half-integer rates	16
Width/rate sweep	Ten widths and five rates at amplitude 0.75	50

G0 is the original profile $G(1, 1/2; x)$ and G2 equals the original narrow Gaussian. The four Gaussian interpolation profiles I0 through I3 follow the order $(5/8, 5/16)$, $(5/8, 7/16)$, $(7/8, 5/16)$, $(7/8, 7/16)$. S0, S1, P0, and P1 follow the order in Section 3. The coefficient-only arm sees only

G0, so its broad “training specifications” row denotes the common evaluation set rather than a set of different ICs that it learned. All other arms train on exactly the same twenty distinct specifications.

C Seed uncertainty and complete error accounting

Table 5: Original-profile relative L2 by rate, mean $\pm$ seed SD. All three conditioned methods satisfy the per-coefficient non-regression tolerance.

ρ	Coefficient only	Raw sensors	Fourier	Reaction Fourier
1	0.0041 ± 0.0022	0.0057 ± 0.0022	0.0043 ± 0.0021	0.0053 ± 0.0011
2	0.0036 ± 0.0022	0.0055 ± 0.0020	0.0043 ± 0.0014	0.0048 ± 0.0013
3	0.0036 ± 0.0020	0.0043 ± 0.0014	0.0035 ± 0.0016	0.0042 ± 0.0007
4	0.0033 ± 0.0020	0.0050 ± 0.0010	0.0030 ± 0.0019	0.0035 ± 0.0012
5	0.0031 ± 0.0019	0.0038 ± 0.0010	0.0030 ± 0.0012	0.0028 ± 0.0011

Table 6: True Euclidean absolute L2 error on the same 1,000 test coordinates per query, mean $\pm$ seed SD. This supplements the original implementation’s MAE in Table 1; it does not relabel MAE as a norm.

Query set	Coefficient only	Raw sensors	Fourier	Reaction Fourier
Original profile	0.091 ± 0.053	0.125 ± 0.037	0.092 ± 0.041	0.106 ± 0.027
Training specifications	8.314 ± 0.057	0.345 ± 0.070	0.385 ± 0.038	0.330 ± 0.096
Gaussian interpolation	7.762 ± 0.062	1.460 ± 0.087	1.229 ± 0.378	0.905 ± 0.124
Sinusoidal profiles	2.473 ± 0.045	1.530 ± 0.131	1.592 ± 0.103	1.257 ± 0.155
Polynomial profiles	2.478 ± 0.051	1.930 ± 0.005	1.558 ± 0.362	1.698 ± 0.407
Recombined pairs	10.122 ± 0.062	2.064 ± 0.461	2.261 ± 1.105	1.736 ± 0.433
Coefficient interpolation	8.640 ± 0.059	1.154 ± 0.434	1.349 ± 0.763	0.927 ± 0.346
Joint interpolation	7.641 ± 0.109	1.553 ± 0.130	1.501 ± 0.662	1.121 ± 0.376
Width/rate sweep	9.945 ± 0.044	5.654 ± 0.566	4.787 ± 0.535	6.339 ± 0.380

Table 7: Paired Reaction Fourier gains against raw sensors. With three seeds, paired t intervals use two degrees of freedom and should be interpreted cautiously. The registered pooled-SD rule in Table 2 is less stringent than requiring a positive 95% interval.

Query set	Paired gain	95% paired t interval	Seeds with positive gain
ic_interpolation	0.0279	[0.0045, 0.0513]	3/3
shape_sinusoid	0.0116	[-0.0115, 0.0348]	3/3
shape_polynomial	0.0115	[-0.0322, 0.0552]	2/3
combination	0.0180	[-0.0145, 0.0506]	3/3
joint_interpolation	0.0243	[-0.0390, 0.0876]	2/3

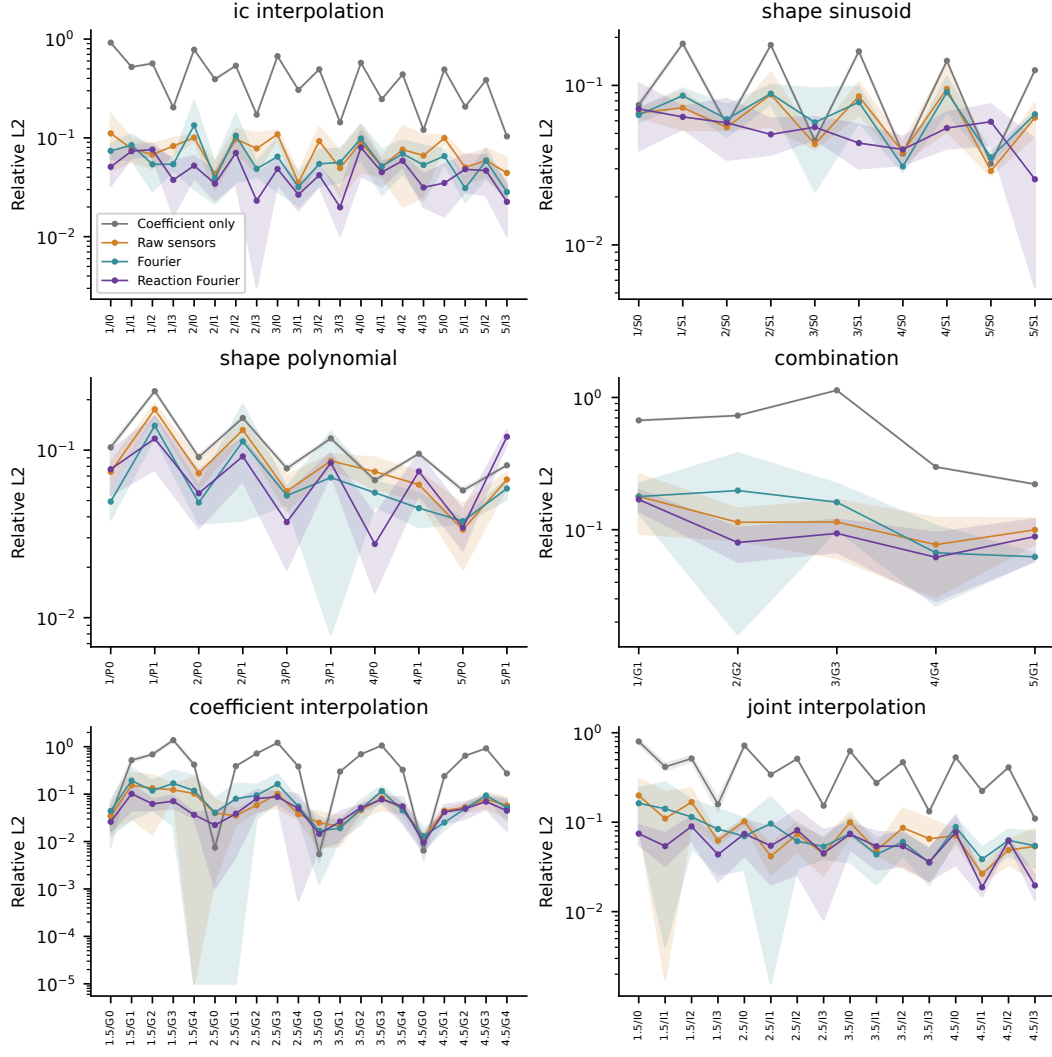


Figure 5: Every individual primary transfer query. The horizontal label is reaction rate/profile identifier. Lines join mean errors for visual comparison, with one seed SD shaded. They are not continuous interpolation paths. Different specifications within an aggregate can have different method rankings.

D Input interventions and training behavior

Table 8: Relative L2 under IC-input interventions, mean $\pm$ seed SD. Coefficients and model weights are unchanged. Constant conditioning supplies G0; shuffled conditioning rotates profile descriptors within each query set. Complete recombination interventions are also retained in the machine-readable logs.

Method	Query set	Correct	Constant profile	Shuffled profile
Raw sensors	Gaussian interpolation	0.074 ± 0.005	0.412 ± 0.004	0.346 ± 0.014
Raw sensors	Sinusoidal profiles	0.064 ± 0.005	0.103 ± 0.002	0.125 ± 0.011
Raw sensors	Polynomial profiles	0.083 ± 0.000	0.106 ± 0.002	0.094 ± 0.005
Raw sensors	Joint interpolation	0.081 ± 0.011	0.397 ± 0.010	0.355 ± 0.014
Fourier	Gaussian interpolation	0.063 ± 0.020	0.413 ± 0.003	0.327 ± 0.023
Fourier	Sinusoidal profiles	0.066 ± 0.004	0.103 ± 0.002	0.098 ± 0.002
Fourier	Polynomial profiles	0.067 ± 0.016	0.106 ± 0.002	0.096 ± 0.009
Fourier	Joint interpolation	0.078 ± 0.036	0.398 ± 0.019	0.328 ± 0.020
Reaction Fourier	Gaussian interpolation	0.046 ± 0.005	0.410 ± 0.005	0.306 ± 0.016
Reaction Fourier	Sinusoidal profiles	0.052 ± 0.006	0.102 ± 0.003	0.123 ± 0.005
Reaction Fourier	Polynomial profiles	0.072 ± 0.018	0.104 ± 0.003	0.116 ± 0.015
Reaction Fourier	Joint interpolation	0.057 ± 0.019	0.405 ± 0.006	0.309 ± 0.014

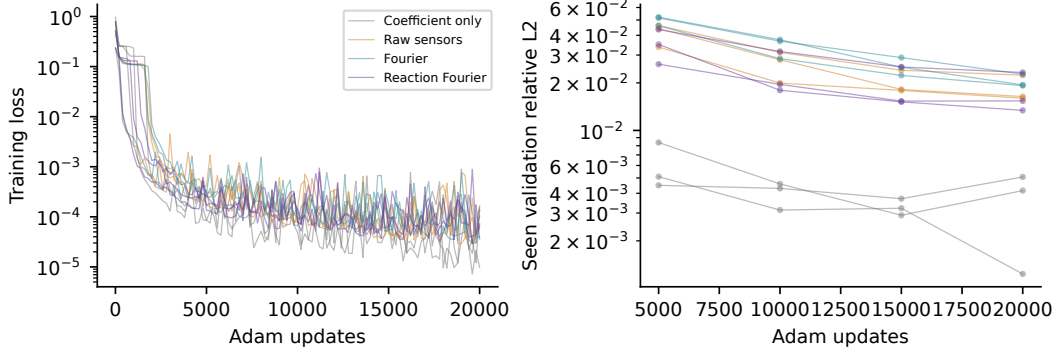


Figure 6: All twelve training histories and seen-specification validation trajectories. Occasional Adam loss excursions persist, motivating reporting of final-epoch seed variation. No curve is used to select a checkpoint. Coefficient-only validation uses its original profile; the other validation curves average their broader trained specification set.

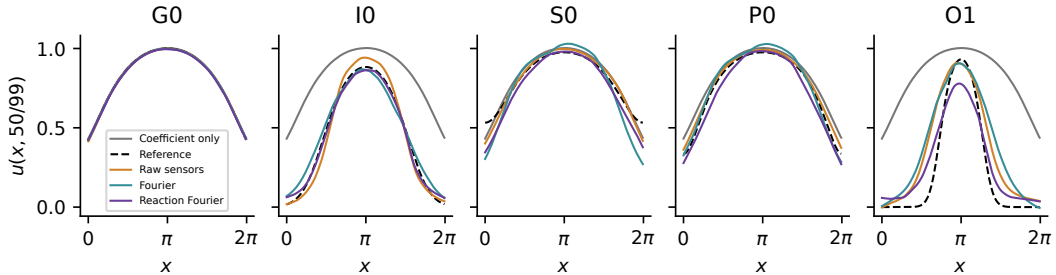


Figure 7: Preselected profile examples at $\rho = 3$, $t = 50/99$, seed 0. The panels show the original Gaussian, a new Gaussian, a sinusoid, a polynomial, and the narrow Gaussian with $w = 0.15$. Examples were fixed before neural test errors were inspected. They illustrate shape-dependent departures from the exact solution rather than selecting only favorable cases.

E Failure analysis and scope of the negative results

Table 9: Representative width/rate-sweep errors, mean $\pm$ seed SD. “Clipping only” is an exploratory reference diagnostic: propagate the pointwise-clipped IC through the same exact logistic construction, then compare against the true reference on the same test points. It is not a lower bound on neural error and is not an alternative reference for scoring models.

ρ	w	Coefficient	Raw	Fourier	Reaction Fourier	Clipping only
3	0.10	2.041 ± 0.006	0.726 ± 0.092	0.710 ± 0.141	0.833 ± 0.235	0.0019
3	0.15	1.516 ± 0.005	0.396 ± 0.054	0.378 ± 0.131	0.420 ± 0.113	0.0011
3	0.25	0.861 ± 0.004	0.098 ± 0.012	0.080 ± 0.036	0.130 ± 0.014	0.0000
3	0.50	0.129 ± 0.003	0.058 ± 0.008	0.063 ± 0.036	0.055 ± 0.012	0.0000
3	0.75	0.124 ± 0.001	0.142 ± 0.075	0.101 ± 0.061	0.135 ± 0.006	0.0000
10	0.10	1.034 ± 0.074	0.590 ± 0.059	0.573 ± 0.053	0.649 ± 0.074	0.3528
10	0.15	0.632 ± 0.057	0.447 ± 0.121	0.396 ± 0.080	0.545 ± 0.091	0.2042
10	0.25	0.247 ± 0.010	0.520 ± 0.092	0.421 ± 0.064	0.619 ± 0.078	0.0000
10	0.50	0.312 ± 0.047	0.514 ± 0.052	0.400 ± 0.060	0.585 ± 0.064	0.0000
10	0.75	0.344 ± 0.047	0.485 ± 0.035	0.395 ± 0.062	0.592 ± 0.055	0.0000

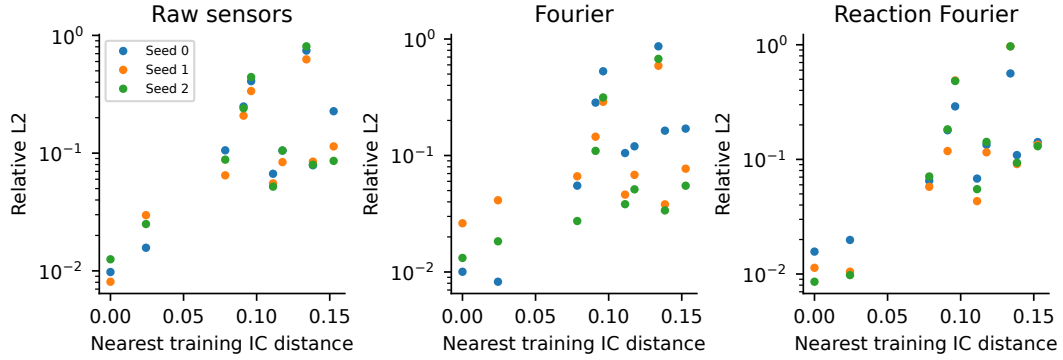


Figure 8: Nearest-profile sensor distance versus width-sweep error at $\rho = 3$. The distance is RMS difference on the common sensors. Its ordering is less closely associated with error than the affine-descriptor-span diagnostic in Figure 3.

The two fully executed structured-encoding attempts are ordinary Fourier and reaction-coordinate Fourier, each at the registered budget and seeds. The latter improves several axes, but neither establishes every registered superiority claim. No seed, profile, or unsuccessful axis is removed. The attempt log records these unresolved comparisons and the numerical diagnostic correction that centers latent vectors in float64; the latter prevents roundoff from assigning spurious nonzero rank to identical coefficient-only codes. Neither correction changes training or model selection.

Boundary-type transfer, an authored non-periodic solver, multi-operator training, and increased-budget boundary discrimination were not part of this registered experiment and are not reported as completed studies. The results do not imply these extensions would fail. They delimit the demonstrated claim: conditional prediction within one nonlinear reaction family, with useful interpolation gains and substantial uncertainty outside the training profile geometry.